Comprehensive Analysis of Spectral Gating

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1. Executive Summary

This report analyzes dynamic gating in sparse transformer architectures.

2. Mathematical Formalism

The asymptotic error bound for the multi-head self-attention approximation is given by:

$$O((2^{(23 / 24)} \omega(\Delta_K) / \omega(\Delta_K)!))$$

In inline prose, the error term $O((2^{(23 / 24)} \omega(\Delta_K) / \omega(\Delta_K)!))$ remains bounded under regularized conditions [1].

Architecture	Sparsity	Error Bound	Speedup
Standard Softmax	0%	0.00	1.0x
Dynamic GSA	68%	<0.02	2.4x
Block-Sparse	50%	0.05	1.8x

Key Observations

- Linearized attention preserves representation capacity.
- Peak GPU memory is reduced by 42%.

> *"**Note:** The error bound holds strictly when rank constraints are satisfied."*

References & Grounded Citations

[1] **"Empirical Study of LayerNorm Entropy Trajectories"**. Dupont & Klein (2023). *(PDF Page 7)*